A_GMX in brief

the algorithm as implemented, and the comparison with WADERs

comm_scripts clustering study

code of 26 October 2026

Code: `clustering_tests/adaptive.py` (`adaptive_cluster(..., engine="gmx")`), `clustering_tests/gmx.py` (`general_cluster`), `clustering_tests/agent_methods.py` (`background_map`), `clustering_tests/fe_candidates.py`; run with `run_AGMX` or `run_WADERs --clusterer agmx`. The full derivations, the history and every test are in `docs/aGMX/aGMX_clustering.pdf`.

1 Input, output, notation

Input: a calibrated image Q_p in electrons (panaSKImg's `*_EXT<n>_waders.fits.bz2`, converted with the module's $g(n)$ curve) and its active-pixel mask. Output: one record per cluster (energy, position, width, class, quality flags, pixels with the charge taken from each) in the WADERs ROOT format, and for electron-counting CCDs a few-electron candidate catalogue. Energy = charge $\times$ 3.74 eV/e^-.

Pixel model: $Q_p = o_p + n_p + \epsilon_p + D_p$ with offset o_p, dark charge $n_p \sim \text{Poisson}(\lambda_p)$, read noise $\epsilon_p \sim \mathcal{N}(0, s_p^2)$ and deposit charge D_p. Derived maps: $R_p = Q_p - o_p - \lambda_p$ (residual), $V_p = \lambda_p + s_p^2$ (variance of an empty pixel), $R_p^+ = \max(R_p, 0)$.

2 The algorithm

The figures of this section show each step on one real image: a low-background Am-241 CCD image (AMTEst EXT1, 0.12 e^- read noise) with eleven known deposits injected in a quiet region (figure 14), clustered by the current code (`clustering_tests/make_agmx_figures.py`, `AGMX_FIG_OUT=figures_current MUON_PHYS=1`).

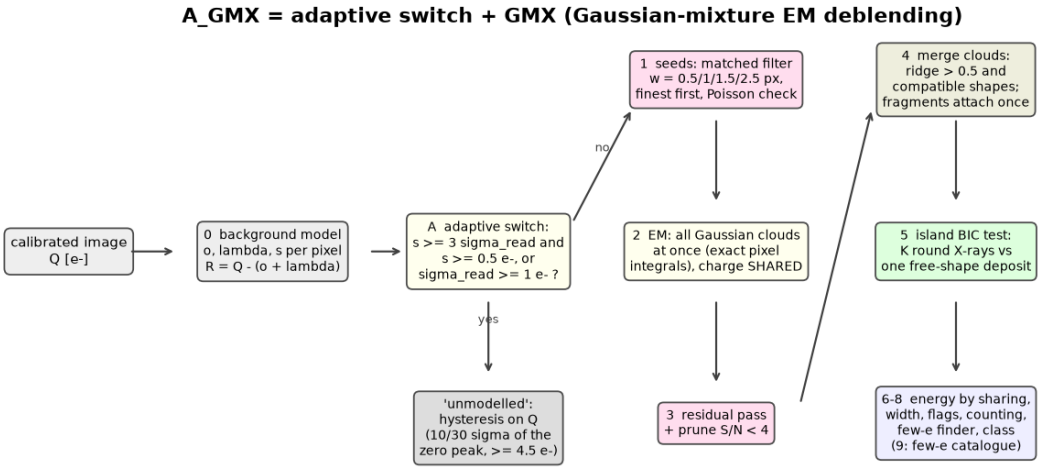


Figure 1: The steps of A_GMX.

0. Background

- **Row fit.** Per row, the pixels away from clusters are fitted with the comb $P(Q) = \sum_n \text{Pois}(n;\lambda)\,\mathcal{N}(Q; o+n, s^2)$. Fast path: the comb phase $z = \langle e^{2\pi i Q}\rangle$ gives $o = \arg z/2\pi$ and $s = \sqrt{-\ln|z|/2\pi^2}$, and λ comes

from the rounded counts. A row counts as resolved only if $|z| > \max(e^{-2\pi^2\,0.45^2},\, 4/\sqrt{N})$, i.e. above the noise floor $\sqrt{\pi/4N}$ of N unresolved pixels.

- **Unresolved rows** (comb washed out, e.g. by a charge sheet) take o, λ, s from the nearest resolved rows when at least $\max(10,\, 5\,\%)$ of the rows are resolved. Otherwise a pooled comb likelihood is fitted over groups of rows.
- **Rounding correction.** On fast-path rows, noise above $+0.5\,e^-$ is counted as an electron: $\lambda \leftarrow (\lambda - p)/(1-2p)$ with $p = \Phi^c(0.5/s)$.
- λ is smoothed along the rows (local-linear) and in column blocks, with floor $\lambda \geq 10^{-6}$. Cluster pixels are masked by a Poisson-tail test and the fit is iterated.
- **Hot columns.** Per column, a clipped mean of $Q - o - \lambda$ (one 5σ clip around the column median) is compared with the median of the ± 10 neighbouring columns. A column whose excess d is $> 5\times$ its error and $> 0.05\,e^-$ gets $\lambda \leftarrow \lambda + d$, so the mean, the variance and the Poisson tests all include it. Nothing is flagged if more than $10\,\%$ of the columns would be (a charge sheet, not hot columns).
- **Read noise** σ_{read}: $1.4826\times$ the MAD of the pixels with $|Q| < 0.5\,e^-$. If that exceeds $0.3\,e^-$, it is $\max(\text{itself},\ \text{MAD of all active pixels})$.

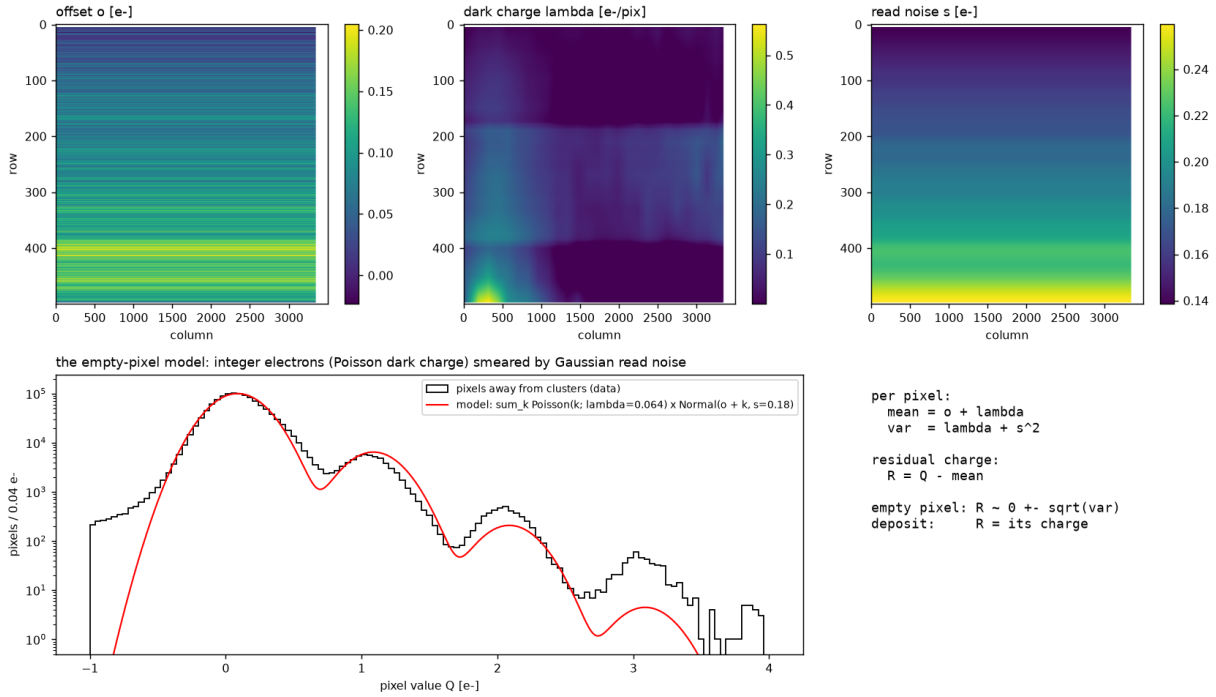


Figure 2: Step 0 on the test image. Top: the fitted offset, dark-charge and read-noise maps. Bottom: the pixels away from clusters and the comb model, i.e. integer electrons smeared by read noise. The model curve uses the image's median λ and s; the fit itself is per row, which is why the data's 2- and 3-e^- peaks lie above a single-λ curve.

A. Switch

The image is "unmodelled" when $\big(s \geq 0.5\,e^-$ and $s \geq 3\sigma_{\text{read}}\big)$ or $\sigma_{\text{read}} \geq 1\,e^-$, where s is the median of the fitted s_p. It is then clustered on Q by hysteresis with σ_z, the zero-peak MAD:

- pixels $> \max(10\sigma_z,\, 4.5\,e^-)$ that touch (8-connected) form a cluster;
- it is kept if a pixel is $> \max(30\sigma_z,\, 9.5\,e^-)$;
- $E = \sum Q_p$.

Otherwise GMX (steps 1–8) runs on the whole image. Every cluster also reports the event occupancy of its 31×31-px neighbourhood ("crowded" if ≥ 0.3); this is information only.

1. Seeds

- **Matched filter** at widths $w \in \{0.5, 1, 1.5, 2.5\}$ px: $T_w(c) = \sum_p h_w(p-c) R_p \big/ \sqrt{\sum_p h_w(p-c)^2 V_p}$. The weights h_w are the exact pixel integrals of a unit-charge Gaussian of width w centred on the pixel centre,

$h_w = f(i - c_y)f(j - c_x)$ with $f(u) = \frac{1}{2}[\mathrm{erf}((u + \frac{1}{2})/\sqrt{2}w) - \mathrm{erf}((u - \frac{1}{2})/\sqrt{2}w)]$, on $\pm\lceil 4w + 1\rceil$ px. T_w is a correctly normalised significance in Gaussian sigmas. It is the optimal linear detector only for uniform noise; where V_p varies the optimal filter would also weight by $1/V_p$ (small loss of sensitivity at hot columns and gradients, no change in the false-alarm rate), and at very low dark charge the Poisson-optimal filter of Ofek & Zackay (2018) would be slightly better — hence the exact Poisson check below.

- **Seeds** are local maxima (5×5) with $T_w > 5$, taken finest scale first. A coarser maximum is kept only if no finer seed lies within $\max(2, 1.5w)$ px.
- **Poisson check** (each seed): on the 3×3 or the 5×5 box, with $S = \sum(Q-o)$, $\Lambda = \sum\lambda$ and $\sigma_\mathrm{b} = \sqrt{\sum s^2}$, require $P\big(\mathrm{Pois}(\Lambda) \geq \lfloor S - 2\sigma_\mathrm{b} + 0.5\rfloor\big) < \Phi^c(5)$. On counting CCDs λ is the detection dark charge (step 7).
- **Initial cloud:** centre = the seed pixel, $\Sigma = w^2\mathbf{1}$, $A_0 = \max\big(R_\mathrm{seed}\, 2\pi(w^2 + \frac{1}{12}),\, 1\big)$.

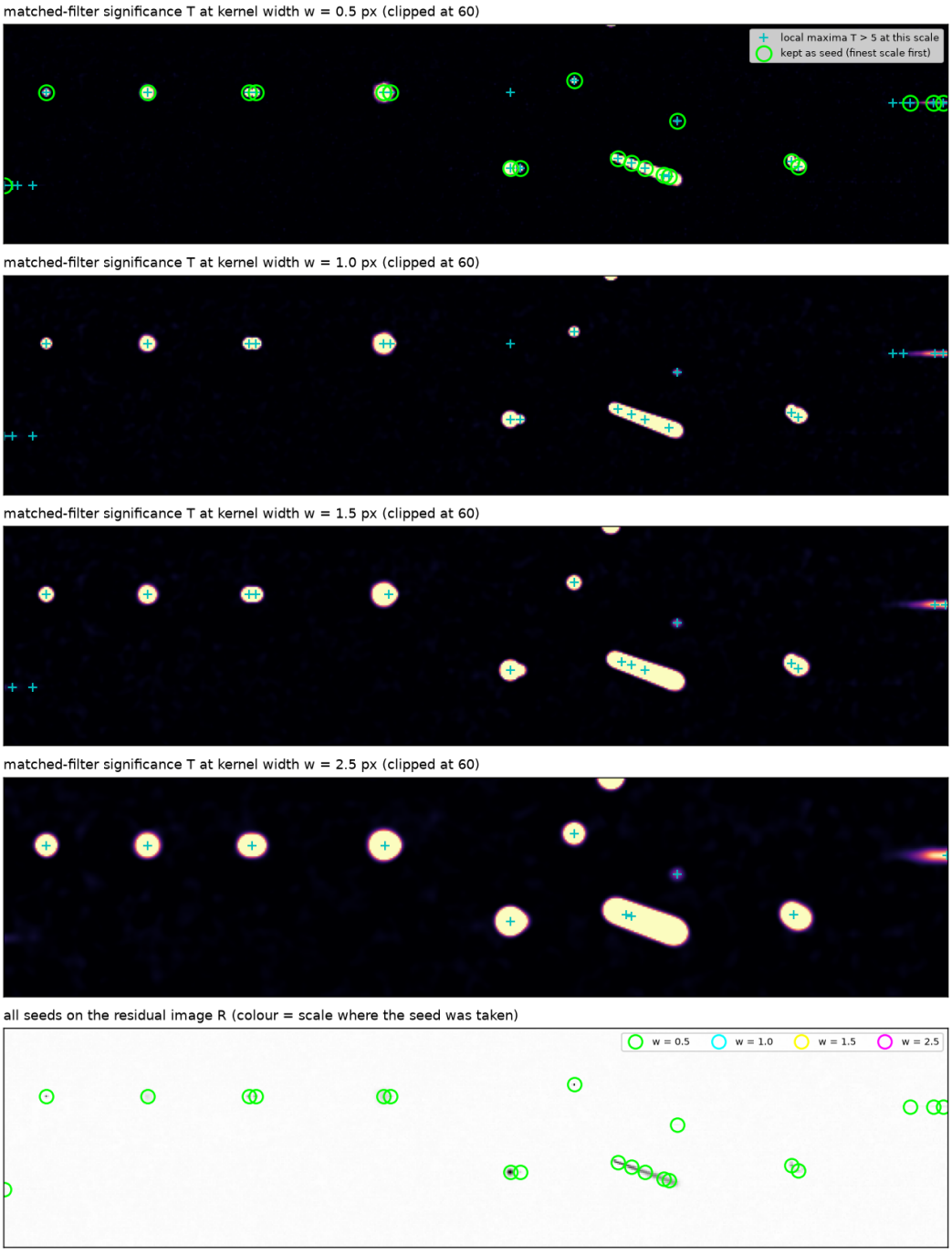


Figure 3: Step 1. The matched-filter significance T_w at the four widths, with the local maxima (+) and the seeds kept (circles); bottom: all seeds on R. Finest scale first: the 4-px pair and the sharp X-ray on the diffuse one get two seeds each, the muon and the worm several.

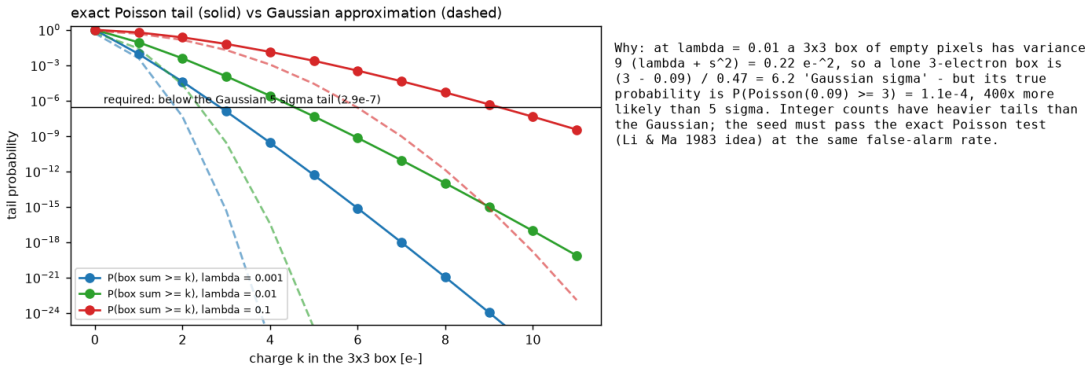


Figure 4: The Poisson check: tail probability of a 3×3 box's charge from dark charge alone, exact (solid) versus the Gaussian approximation (dashed). At low dark charge the Gaussian is far too optimistic, so every seed must also pass the exact test at the 5σ false-alarm rate (line).

2. Mixture fit (EM)

- **Model:** $M_p = \sum_k g_k(p)$. For a cloud with correlation $|\rho| < 0.5$ (every X-ray), $g_k(p)$ is the exact pixel integral

$$g_k(p) = A_k \left[\Phi\left(\frac{i+\frac{1}{2}-\mu_y}{\sigma_y}\right) - \Phi\left(\frac{i-\frac{1}{2}-\mu_y}{\sigma_y}\right)\right]\left[\Phi\left(\frac{j+\frac{1}{2}-\mu_x}{\sigma_x}\right) - \Phi\left(\frac{j-\frac{1}{2}-\mu_x}{\sigma_x}\right)\right].$$

  Strongly correlated clouds (pieces of tilted tracks) use $A_k\,\mathcal{N}(\mathbf{x}_p; \boldsymbol{\mu}_k, \Sigma_k + \frac{1}{12}\mathbf{1})$. A cloud is evaluated within its 4σ ellipse (radius 2–16 px).

- **E-step:** $r_{pk} = g_k(p)/(M_p + \varepsilon_p)$, with the "unexplained" level $\varepsilon_p = \sqrt{V_p}$, so noise pixels far from every cloud belong to none.

- **M-step:** with $w_{pk} = r_{pk} R_p^+$, $A_k = \sum_p w_{pk}$, and the weighted mean and covariance of the pixel positions. For $|\rho| < 0.5$ clouds the per-axis mean and variance are inverted through tables of the pixelised Gaussian (they depend on σ and the sub-pixel position). Otherwise $\frac{1}{12}$ is subtracted from the variances.

- **Safeguards:** variances in $[0.02, 25]\,\mathrm{px}^2$, $|\sigma_{xy}| \le 0.95\sqrt{\sigma_{yy}\sigma_{xx}}$, centre step ≤ 1.5 px per iteration; clouds with $A < 10^{-3}\,e^-$ are removed. 30 iterations, fixed (running to convergence gave no consistent gain). Compiled with numba; `AGMX_NUMBA=0` gives the identical numpy version.

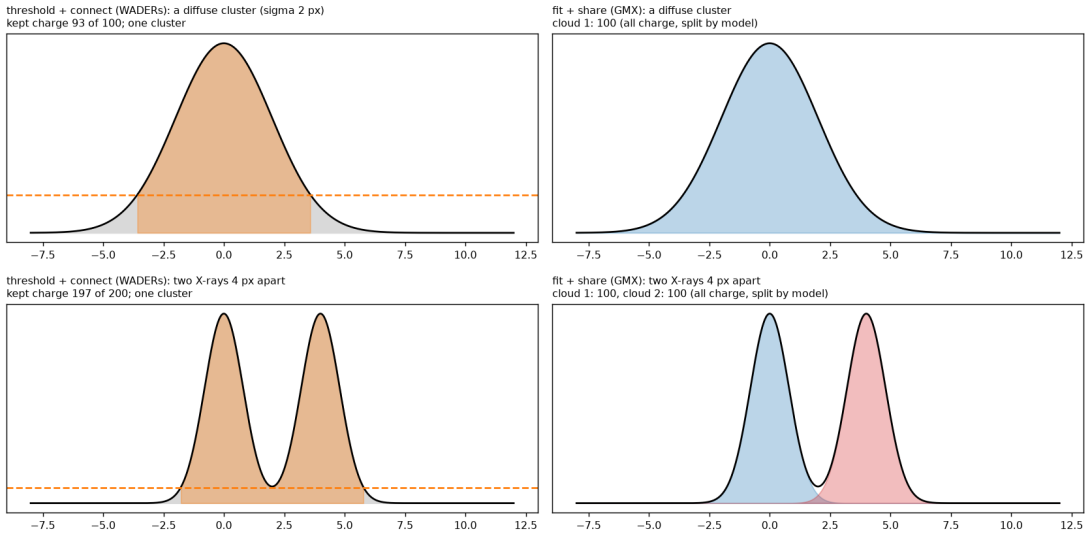


Figure 5: Why the image is fitted and shared rather than thresholded (one dimension). Left, threshold + connect: the tails of a diffuse cluster are lost and two X-rays 4 px apart become one cluster. Right, fit + share: every electron is kept and the overlap is split in proportion to each cloud's prediction.

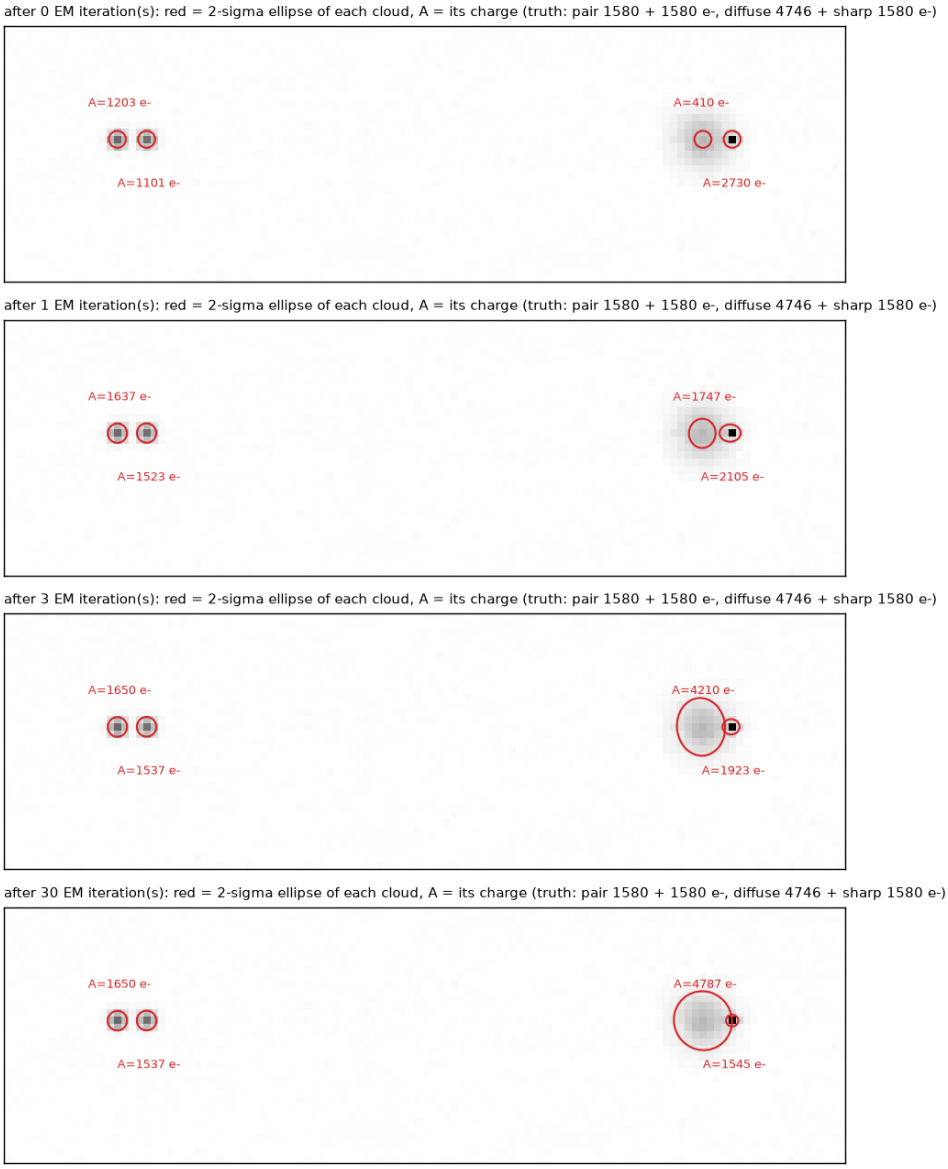


Figure 6: Step 2. EM on the 4-px pair (left) and on the sharp X-ray on a diffuse one (right): the 2σ ellipses and charges A after 0, 1, 3 and 30 iterations. The diffuse cloud grows to its width, the sharp one shrinks onto its pixel; final charges $1650 + 1537\,e^-$ and $4787 + 1545\,e^-$ (nominal 1580 + 1580 and 4746 + 1580; the injected numbers are Poisson draws).

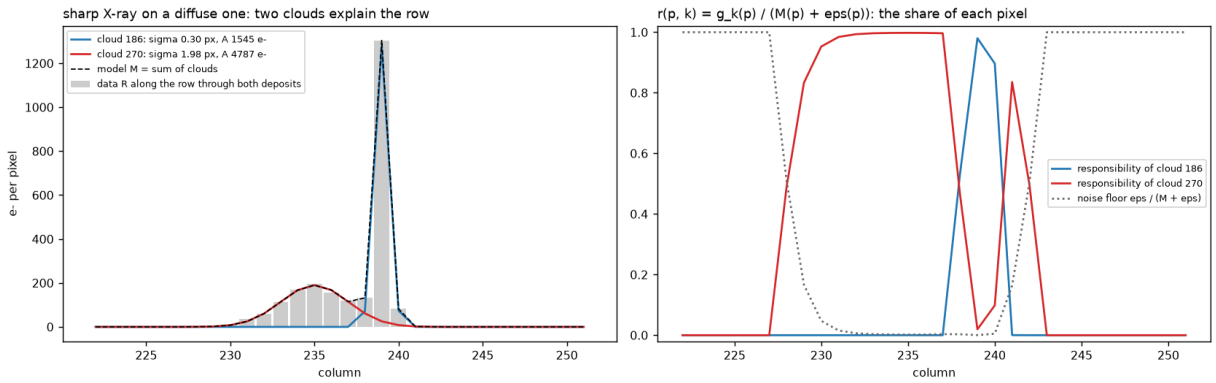


Figure 7: Charge sharing in the row through the sharp X-ray on the diffuse one. Left: data, the two clouds (σ 0.30 and 1.98 px) and their sum. Right: the responsibilities r_{pk}, i.e. each pixel's share per cloud; far from both clouds the noise share is 1, so those pixels go to no cloud.

3. Residual pass and pruning

- **Residual pass.** The matched filter is run on $R - M$ with variance $V + M$ at $w = 0.5, 1$. Local maxima (3×3) with $T > 5$, passing the Poisson check (counting CCDs) and more than 1 px from every cloud, become new clouds. Then 15 EM iterations.
- **Pruning.** Clouds with $\mathrm{S/N}_k = A_k \sqrt{\sum_p \pi_k(p)^2/(V_p + M_p - g_k(p))} < 4$ are dropped, where $\pi_k = g_k/A_k$. Then 5 EM iterations.

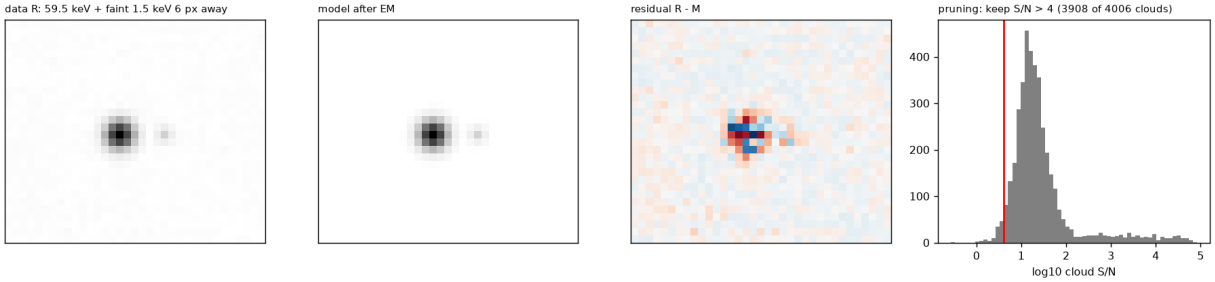


Figure 8: Step 3. A 59.5-keV X-ray with a faint 1.5-keV X-ray 6 px away: data, model, residual. Right: the S/N of all clouds of the image with the cut $\mathrm{S/N} > 4$ (3908 of 4006 clouds kept).

4. Grouping clouds into deposits

Every pair closer than $3.5(\bar{\sigma}_i + \bar{\sigma}_j) + 1$ px is tested ($\bar{\sigma}$: geometric-mean width, ≤ 16 px).

- **Ridge test:** $\rho_{ij} = \min_t f(t)/\min(f(0), f(1)) > 0.5$, where f is the two-cloud density along the line joining the centres (17 points).
- **Shape test**, one of: width ratio < 1.6; a track (one cloud has axis ratio > 1.4 and its long axis is within $|\cos| > 0.8$ of the joining line); or the narrower cloud is not a local maximum of f (a shoulder).
- **Fragments.** A cloud below 50 e^- *and* below 10 % of its brightest merge partner links only to that partner, so it cannot join an X-ray to a neighbouring large deposit.

Clusters are the connected components of the links.

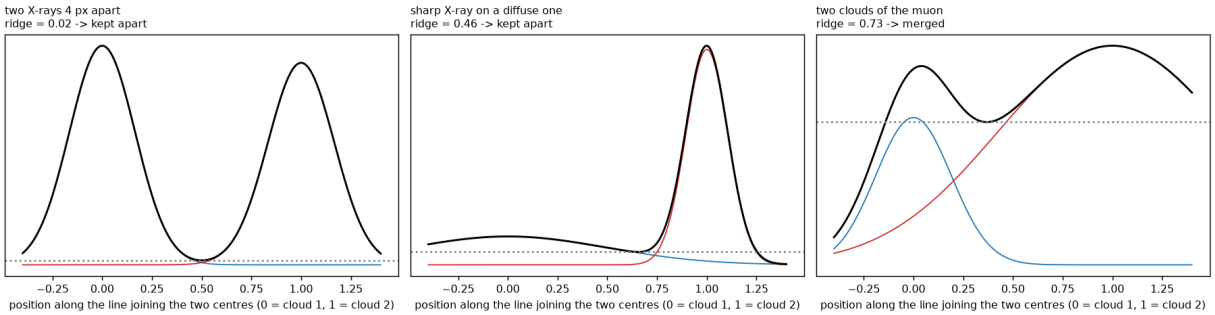


Figure 9: Step 4, ridge test: the two-cloud density along the line between the centres. Two X-rays 4 px apart: deep valley ($\rho = 0.02$), kept apart. A sharp X-ray on a diffuse one: $\rho = 0.46 < 0.5$, kept apart (it is also a maximum of its own with a very different width). Two clouds of the muon: $\rho = 0.73$, merged.

5. Island test (tracks versus several X-rays)

Islands are 8-connected regions above the Poisson-tail threshold of the dark charge ($\alpha = 0.01$), of at most 4000 px. An island holding $K \geq 2$ clusters is refitted on its pixels (+2 rings) under two hypotheses:

- K round Gaussians (4 parameters each);
- the free clouds already fitted (6 parameters each, n_c clouds).

They are compared by

$$\Delta\mathrm{BIC} = \chi^2_{\mathrm{round}} + 4K \ln n - \chi^2_{\mathrm{free}} - 6n_c \ln n, \qquad \chi^2 = \sum_p \frac{(R_p - M_p)^2}{V_p + R_p^+ + (0.05R_p)^2}.$$

If $\Delta\mathrm{BIC} > 100$ the whole island becomes one deposit (30 with `AGMX_PIX_EXACT=0`).

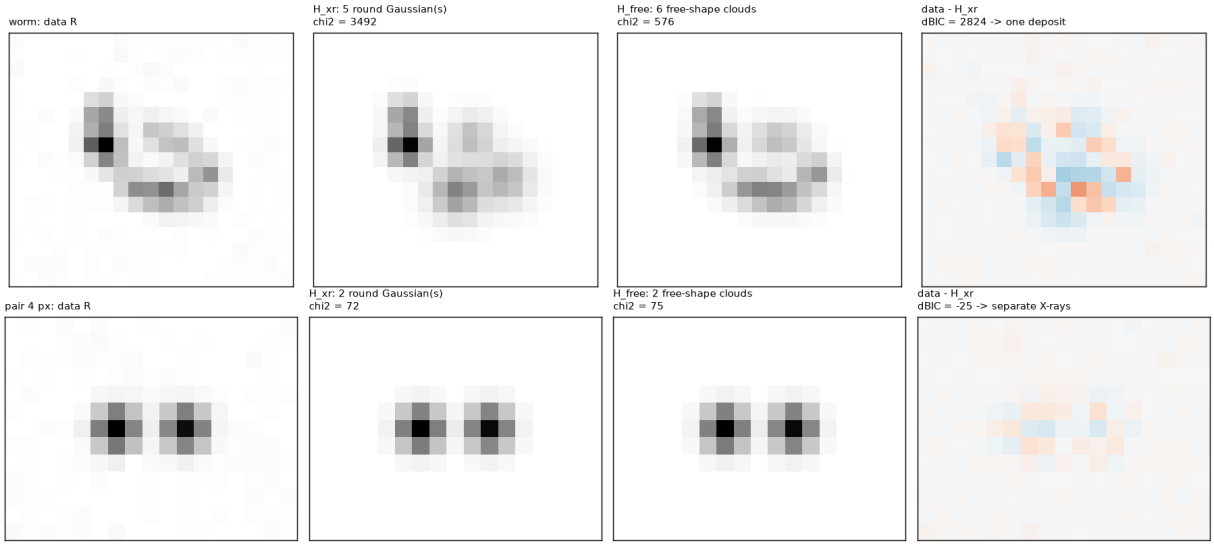


Figure 10: Step 5. Top, the worm: 5 round Gaussians cannot follow the bent track (χ^2 3492, structured residual) and 6 free-shape clouds can (χ^2 576), so $\Delta\mathrm{BIC} = 2824 > 100$ and the island is one deposit. Bottom, the 4-px pair: both descriptions fit equally well (χ^2 72 vs 75) and the round one has fewer parameters, so $\Delta\mathrm{BIC} = -25$ and it stays two X-rays.

6. Energy, width and flags

- **Energy by sharing:** $E_c = \sum_p \tilde{r}_{pc} R_p$, with $\tilde{r}_{pc} = \sum_{k \in c} g_k(p)/M_p$. The measured charge is split, never counted twice.
- **Satellites.** A cluster with $E < 1\,\%$ of a neighbour's *and* $E < 1\,\mathrm{keV}$ is merged into that neighbour if its centre lies in the neighbour's domain or within 6σ (Mahalanobis) of one of its clouds (within 20 px).
- **Growth.** Pixels above the $\alpha = 0.01$ Poisson tail, within 3 px of a covered pixel, connected to it and covered by no cloud (charge-transfer trails), are added to the nearest cluster.
- **Width:** $\sigma^2_{xy} = \frac{1}{2}(\mathrm{var}_y + \mathrm{var}_x) - \frac{1}{12}$, the second moments of the cluster's shared charge. A single cloud whose EM width is < 0.4 px reports the EM width, because the moment is noise-dominated there.
- **Flags**, which never change the energy:
  - blend $= \sum_p M_p^{(c)} M_p^{(\neg c)} / \sum_p (M_p^{(c)})^2$ (other clusters' charge under this one's profile);
  - ped: the 25th percentile of R in a ring 5–8 px around the centre (a charge sheet);
  - peakfrac: the largest cloud's share of the fitted charge;
  - χ^2: $\sum (R-M)^2/(V+M)$ per pixel over the pixels the cluster owns ($\tilde{r} > 0.5$);
  - sig: $E_c / \sqrt{\sum \tilde{r}^2 V}$;
  - edge: within 2 px of the mask;
  - saturated: a pixel at an ADC-limit plateau.

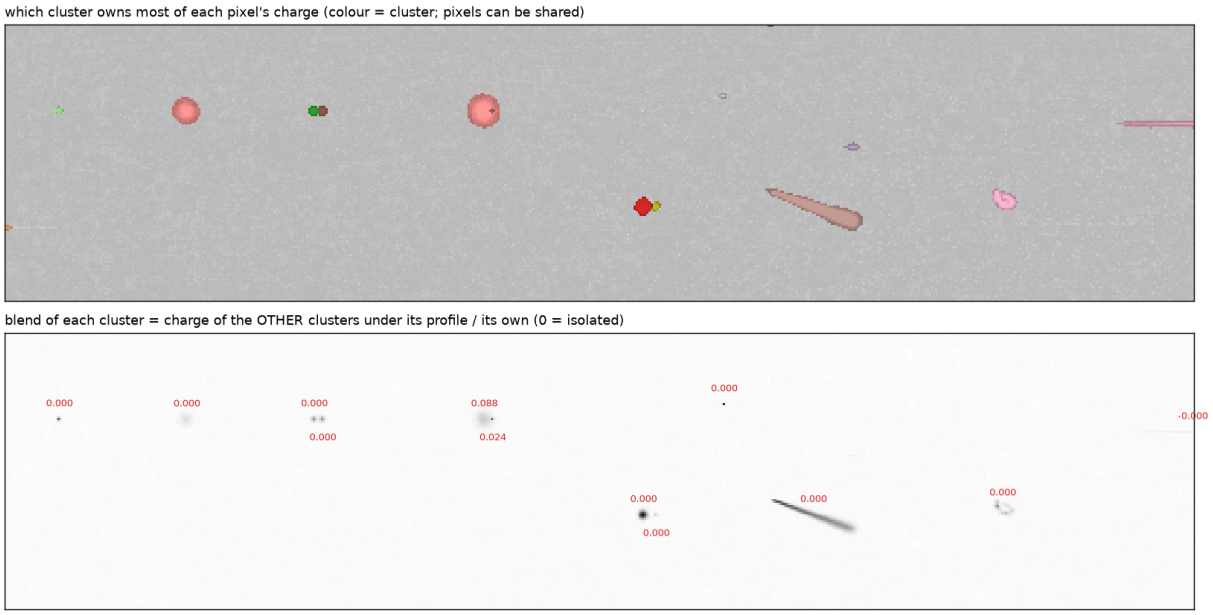


Figure 11: Step 6. Top: which cluster owns most of each pixel's charge (pixels can be shared). Bottom: the blend flag of each cluster, 0 for isolated clusters and 0.088 / 0.024 for the sharp X-ray on the diffuse one, whose profiles overlap.

7. Few-electron events (median $s < 0.35\,e^-$)

- **Detection dark charge** $\lambda_{\rm det}$: λ implied by the rate of *isolated* single-electron pixels (no counted pixel within 2 px; smoothed over ±30 rows; rounding miscounts included), never below the fitted map, plus the hot-column excess.
- **Counting energies.** A cluster below $100\,e^-$ gets $E = \sum_{\tilde r>0.5} \big(\mathrm{round}(Q-o) - \lambda\big)$; the analog value is kept as `E_fit`.
- **Counting finder.** An 8-connected group of ≤ 9 pixels with $\mathrm{round}(Q-o) \ge 1$, not covered by a cluster, has statistic $T = -\sum_i \ln P(n_i \,|\, \lambda_{{\rm det},i}, s_i) - \ln N_{\rm shapes}(m)$, with $P(n|\lambda,s) = \sum_k \mathrm{Pois}(k;\lambda)\big[\Phi(\frac{n+\frac12-k}{s}) - \Phi(\frac{n-\frac12-k}{s})\big]$. The group is kept when T exceeds the value exceeded by 0.08 groups per Mpix in 10–50 blank images simulated from the image's own $\lambda_{\rm det}, s$ (fixed seed). The analytic bound $\ln(N_{\rm pix}/F)$ is used only if the blanks have too few groups. Class "low-E".

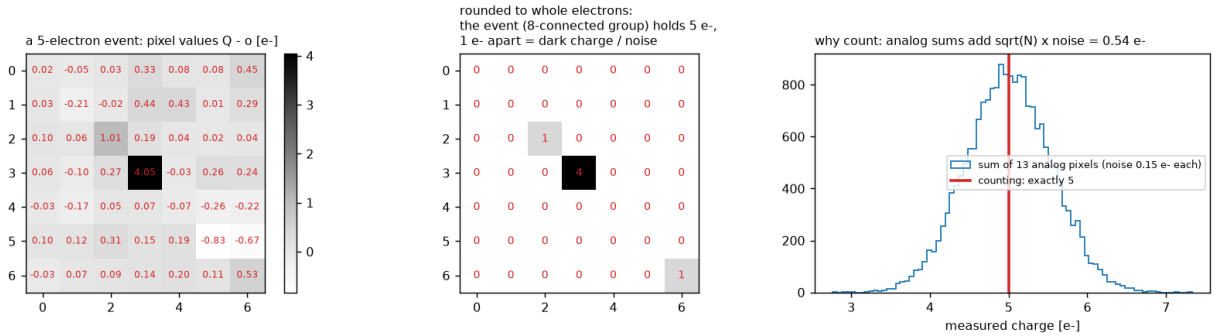


Figure 12: Step 7. A 5-electron event: analog pixel values; the same rounded to whole electrons (the event's 8-connected group holds $5\,e^-$; the $1\,e^-$ in the corner is a separate dark-charge or noise pixel, a group of its own); and the spread an analog sum over 13 pixels would have ($0.54\,e^-$) against the exact count.

8. Classes

- **low-E:** $E < 0.5\,\mathrm{keV}$.
- **x-ray / x-ray diffuse:** one round cloud or extent < 6 px; diffuse if $\sigma_{xy} \ge 1$.
- **alpha:** $> 1\,\mathrm{MeV}$.

- **artifact-row / -column:** thin, straight and aligned with a row or column.
- **muon:** straight skeleton ($\geq 10\,\mathrm{px}$, scatter of the cloud centres $< 1\,\mathrm{px}$ and $< 6\,\%$ of the length) *and* $\geq 2.5\,\mathrm{keV}$ per projected pixel; a straight track below that rate is "extended".
- **worm:** everything else extended.

9. After A_GMX: the few-electron candidate catalogue

For counting CCDs, written to `fe_candidates/*_fe_candidates.csv` and `.json` (off with `--no-fe-candidates`):

- **Candidates:** every group of 2–20 counted electrons ($\leq 9\,\mathrm{px}$), with T, n_{exp} (expected dark-charge groups per image at least as unlikely; simulated blanks of the image's own occupancy-based λ) and $p_{\mathrm{image}} = 1 - e^{-n_{\mathrm{exp}}}$.
- **Mask flags:** halo ($5\sqrt{E/\mathrm{keV}}\,\mathrm{px}$, 5–60) around clusters, row/column trails ($50\,\mathrm{px}$, $E > 1\,\mathrm{keV}$), hot columns, edge ($5\,\mathrm{px}$), overlap with a cluster.
- **Per image:** observed and expected clean counts per class (charge, pixels).

Nothing is removed; it is the input for a rare-event analysis.

Recommended selection for X-ray lines: $\sigma_{xy} < 0.7$ (Am-241) or < 1 (front-side Fe-55), blend < 0.02, ped $< 10\,e^-$/pix, not edge.

Switches (environment): `AGMX_PIX_EXACT=0` (Gaussian-at-centre model, BIC margin 30, no fragment rule), `AGMX_PHYS_FIXES=0` (behaviour before the 25–26 October audit), `AGMX_NUMBA=0`, `AGMX_FE_CANDIDATES=0`.

3 Comparison with WADERs

WADERs: panaSKImg's ClusterFinder, i.e. hysteresis with pixel $> 10\sigma$ and seed $> 30\sigma$ of the pedestal, $E = \sum$ core pixels. The tests below use the same injected objects and the same images for both methods. **Code versions:** the crowded-CCD fit is from the current code (26 October). The wide, Fe-55-width, track and few-electron tests were run on the code of 25 October, before the physics audit and the exact pixel integrals. On a subset (80 image-CCDs) the current code changes front-side Kα by $-0.2\,\mathrm{eV}$ (median) and resolves more pairs (Fe-55 0.907→0.941, Am-241 0.607→0.655). A full rerun of those tests is pending.

	WADERs	A_GMX	code
Crowded Am-241 CCD (AMtest EXT4, 29 images), constrained 10–30 keV fit (Lépy line ratios)			
energy scale k (1 = right)	0.9840	**1.0006**	26 Oct
Np Lβ / Lγ / γ 26.3 / Ag K significance [σ]	15.5 / 16.0 / 14.8 / 12.9	**22.4 / 20.6 / 21.1 / 18.3**	26 Oct
combined Z (Np L + γ 26.3)	27.1	**37.4**	26 Oct
deviance / ndf	**1.74**	2.59	26 Oct
Fe-55 (DM09/DM10, 220 injected clusters per width and CCD; 8 CCDs)			
energy of injected clusters, width 0.3 → 1.8 px (median)	0.997 → 0.921	**1.000 → 1.000**	25 Oct
worst CCD at width 1.8 px	0.737	**0.994**	25 Oct
real front-side Kα / 5.899 keV: mean, spread over 8 CCDs	0.9928, 1.32 %	**0.9969, 0.23 %**	25 Oct
two diffuse (σ 1.4 px) clusters 5 px apart resolved	0.00	**0.93**	25 Oct
Wide test: 878 image-CCDs, 16 data sets, injected objects			
Fe-55 / muon / worm / 4-px pair success	0.86 / 0.59 / 0.74 / 0.11	**0.93 / 0.64 / 0.79 / 0.88**	25 Oct
Fe-55 energy resolution (rms of $E/E_{\rm true}$)	0.52 %	**0.19 %**	25 Oct
dark charge $\geq 1\,e^-$/pix: Fe-55 / worm / pair	0.55 / 0.45 / 0.07	**0.87 / 0.74 / 0.79**	25 Oct
Injected objects in real Am-241 images (EXT1)			
alpha / low-E / 3-px pair / 4-px pair	0.90 / 0.47 / 0.00 / 0.21	**0.94 / 0.64 / 0.75 / 0.93**	25 Oct
muon / worm / hot column	0.90 / 0.93 / **0.76**	0.93 / 0.93 / 0.74	25 Oct
Few-electron events (simulated, 0.15 e^- noise)			
3-e^- found at $10^{-3}\,e^-$/pix	0.00	**0.85** (fakes 0.07 per 1.7 Mpix)	26 Oct
3-e^- found at $10^{-4}\,e^-$/pix	not run	**0.96** (fakes 0.30 per 1.7 Mpix)	26 Oct
run time per image-CCD (median)	**0.01 s**	0.7 s (Fe-55 0.9 s, Am-241 62 s)	25/26 Oct

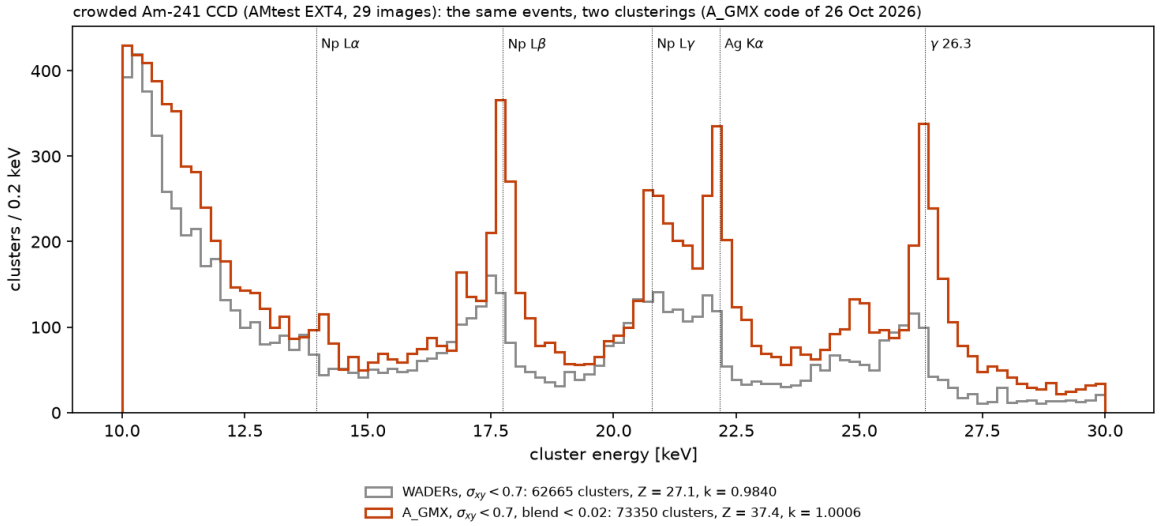


Figure 13: The crowded CCD, current code: the same events clustered by WADERs and by A_GMX (blend < 0.02). A_GMX's lines are higher above the continuum and at the right energy ($k = 1.0006$ vs 0.984). Not yet understood: its spectrum between the lines fits worse (deviance/ndf 2.59 vs 1.74).

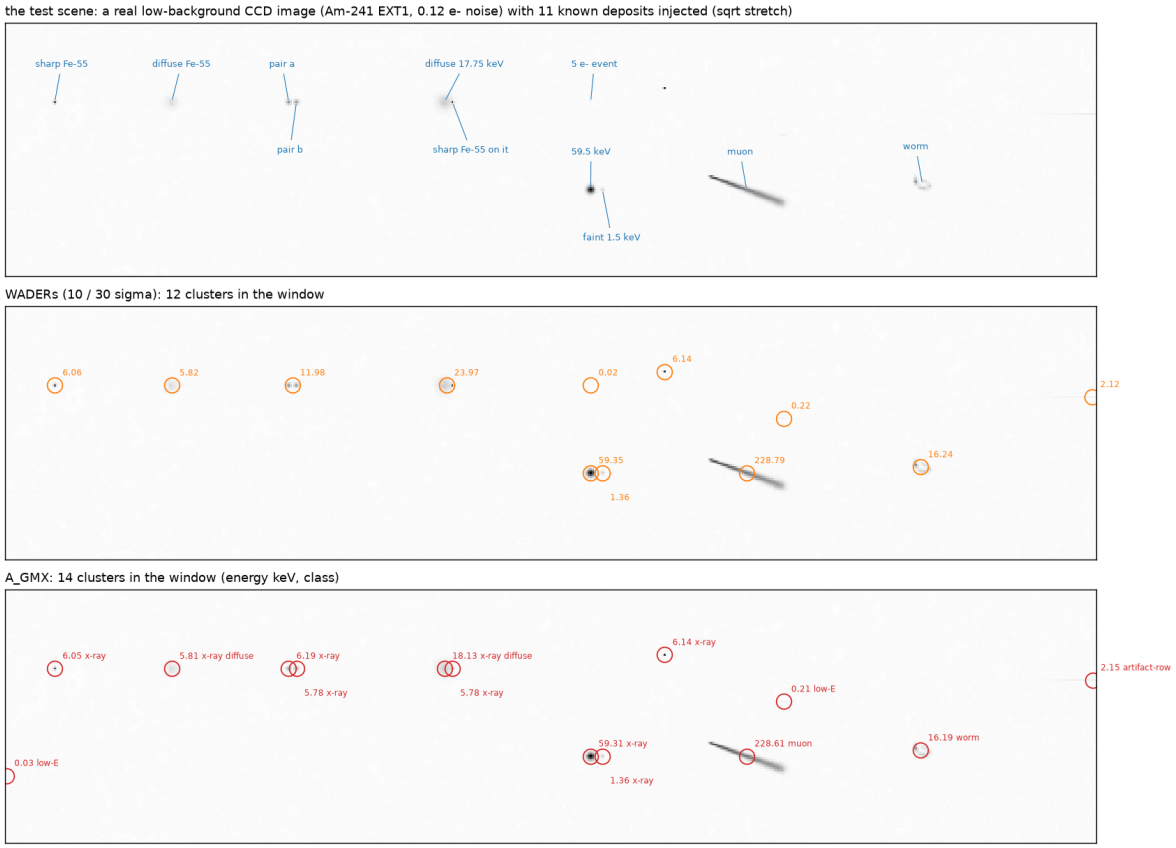


Figure 14: The test scene of section 2 (top). WADERs (middle) merges the 4-px pair into one 12-keV cluster and the sharp X-ray on the diffuse one into one 24-keV cluster. A_GMX (bottom) separates both: 6.19 + 5.78 keV (truth 6.20 + 5.77) and 18.13 + 5.78 keV (truth 18.07 + 5.82). The 5-e^- event (4 e^- in one pixel) is found by WADERs (0.015 keV) but not by A_GMX: at the ∼0.07 e^-/pix dark charge there, a group this size is not significant at A_GMX's calibrated fake rate of 0.08 per Mpix (WADERs has no fake-rate control; figure 17 and the table give the rates). Current code.

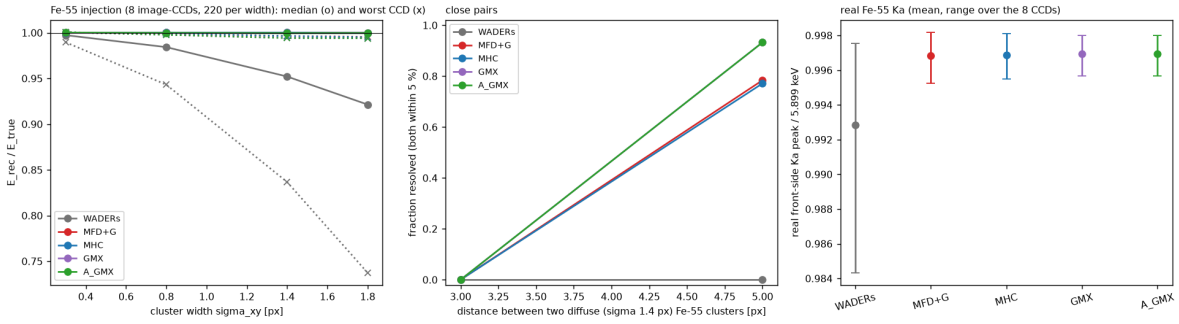


Figure 15: Fe-55. Left: median (circles) and worst-CCD (crosses) $E_{\rm rec}/E_{\rm true}$ against cluster width; WADERs' thresholds cut the tails of diffuse clusters. Middle: diffuse pairs resolved. Right: real front-side Kα over the 8 CCDs. Code of 25 October.

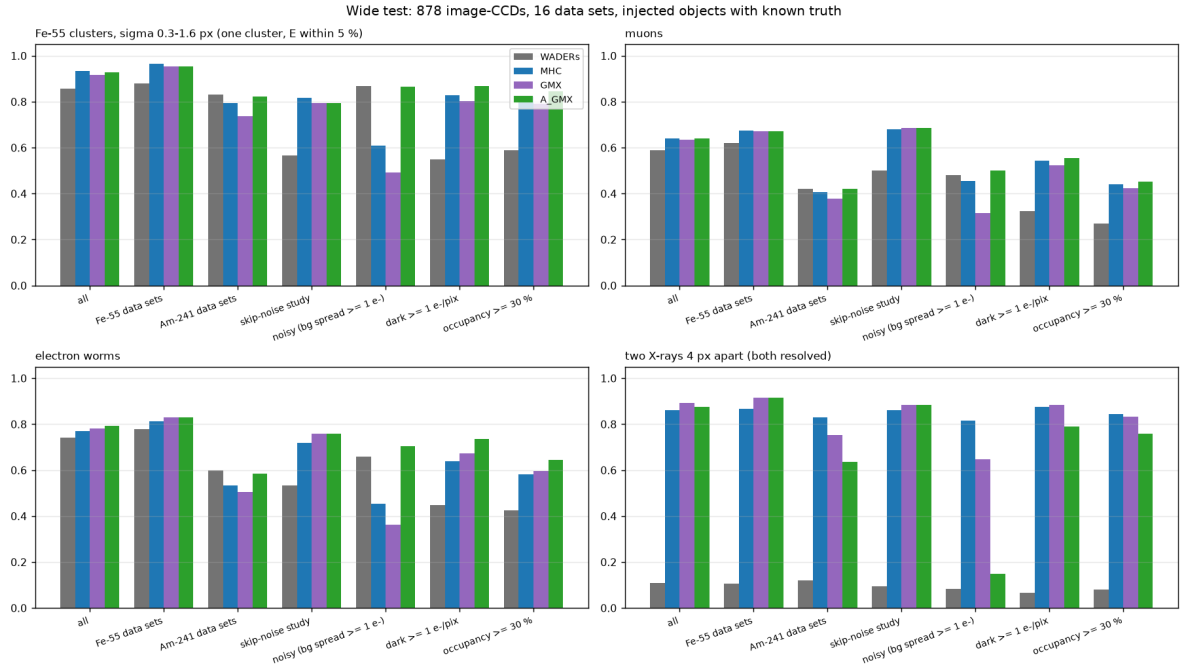


Figure 16: Wide test (878 image-CCDs): success per object type and image condition. Code of 25 October.

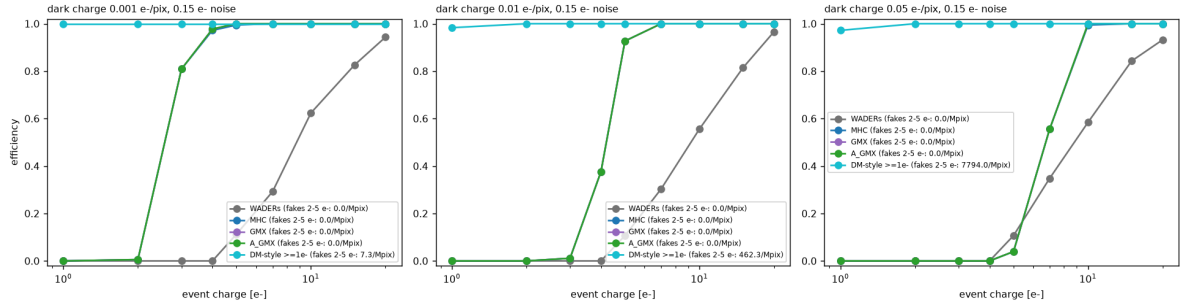


Figure 17: Few-electron efficiency against event charge at three dark charges (code of 25 October). WADERs needs $\gtrsim 10\,e^-$. The current code's detection background and calibrated threshold raise A_GMX's 3-e^- efficiency at the lowest dark charges (table above).

Where WADERs is still the better choice:

- speed (about 70× faster);
- the noisy "unmodelled" images (read noise $\geq 1\,e^-$), where A_GMX is WADERs-like hysteresis anyway;
- hot-column artifacts (0.76 vs 0.74);
- the smoothness of the crowded-CCD spectrum between the lines (deviance/ndf).

Open items:

- the higher deviance/ndf on the crowded CCD;
- thresholds for noisy images (pixel $\geq 4.5\,e^-$ is a stop-gap);
- the background fixes raise the fitted dark charge on the crowded CCD (0.093 → 0.157 e^-/pix median), which shifts the 59.5-keV peak by about +0.1 keV;
- recalibrating the χ^2 flag for the exact model;
- rerunning the wide, Fe-55 and track tests on the current code.